Topic: Initial value problems with nonhomogeneous equations

Question: Solve the initial value problem for the second order nonhomogeneous equation, given $y(0) = 0$ and $y'(0) = 0$.

$$y'' + 2y' + y = 4\sin x$$

Answer choices:

A $y(x) = 2xe^{-x} - 2\sin x$

B $y(x) = 2e^{-x} + 2xe^{-x} - 2\cos x$

C $y(x) = 2e^{-x} - 2\cos x$

D $y(x) = 2e^{-x} + 2xe^{-x} - 2\sin x$

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Solution: B

First we solve the associated homogeneous equation.

$$y'' + 2y' + y = 0$$

$$r^2 + 2r + 1 = 0$$

$$(r+1)^2 = 0$$

$$r = -1, -1$$

So the complementary solution with equal real roots is

$$y_c(x) = c_1 e^{-x} + c_2 x e^{-x}$$

Our guess for the particular solution will be

$$y_p(x) = A\cos x + B\sin x$$

$$y_p'(x) = -A\sin x + B\cos x$$

$$y_p''(x) = -A\cos x - B\sin x$$

Substituting into the original differential equation gives

$$-A\cos x - B\sin x + 2(-A\sin x + B\cos x) + A\cos x + B\sin x = 4\sin x$$

$$-A\cos x - B\sin x - 2A\sin x + 2B\cos x + A\cos x + B\sin x = 4\sin x$$

$$2B\cos x - 2A\sin x = 4\sin x$$

Equating coefficients gives $A = -2$ and $B = 0$, and therefore

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$$y_p(x) = -2\cos x$$

The general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1 e^{-x} + c_2 x e^{-x} - 2\cos x$$

Differentiating the general solution gives

$$y'(x) = -c_1 e^{-x} + c_2 e^{-x} - c_2 x e^{-x} + 2\sin x$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 0$ into the general solution and its derivative.

$$0 = c_1 + 0 - 2$$

$$0 = -c_1 + c_2 - 0 + 0$$

Simplifying these equations gives

$$c_1 = 2$$

$$c_2 = c_1 = 2$$

With $c_1 = 2$ and $c_2 = 2$, the general solution becomes

$$y(x) = 2e^{-x} + 2xe^{-x} - 2\cos x$$

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Topic: Initial value problems with nonhomogeneous equations

Question: Solve the initial value problem for the second order nonhomogeneous equation, given $y(0) = 0$ and $y'(0) = 0$.

$$y'' - 6y' + 5y = 6e^{2x} + 4e^{3x}$$

Answer choices:

A $y(x) = 2e^{x} - 2e^{2x} - e^{3x} + e^{5x}$

B $y(x) = -2e^{x} + 2e^{2x} + e^{3x} - e^{5x}$

C $y(x) = e^{x} - e^{2x} - 2e^{3x} + 2e^{5x}$

D $y(x) = -e^{x} + e^{2x} + 2e^{3x} - 2e^{5x}$

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Solution: A

First we solve the associated homogeneous equation.

$$y'' - 6y' + 5y = 0$$

$$r^2 - 6r + 5 = 0$$

$$(r - 1)(r - 5) = 0$$

$$r = 1, 5$$

So the complementary solution with distinct real roots is

$$y_c(x) = c_1 e^x + c_2 e^{5x}$$

Our guess for the particular solution will be

$$y_p(x) = Ae^{2x} + Be^{3x}$$

$$y_p'(x) = 2Ae^{2x} + 3Be^{3x}$$

$$y_p''(x) = 4Ae^{2x} + 9Be^{3x}$$

Substituting into the original differential equation gives

$$4Ae^{2x} + 9Be^{3x} - 6(2Ae^{2x} + 3Be^{3x}) + 5(Ae^{2x} + Be^{3x}) = 6e^{2x} + 4e^{3x}$$

$$4Ae^{2x} + 9Be^{3x} - 12Ae^{2x} - 18Be^{3x} + 5Ae^{2x} + 5Be^{3x} = 6e^{2x} + 4e^{3x}$$

$$-3Ae^{2x} - 4Be^{3x} = 6e^{2x} + 4e^{3x}$$

Equating coefficients gives $A = -2$ and $B = -1$, and therefore

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$$y_p(x) = -2e^{2x} - e^{3x}$$

The general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1e^x + c_2e^{5x} - 2e^{2x} - e^{3x}$$

Differentiating the general solution gives

$$y'(x) = c_1e^x + 5c_2e^{5x} - 4e^{2x} - 3e^{3x}$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 0$ into the general solution and its derivative.

$$0 = c_1 + c_2 - 2 - 1$$

$$0 = c_1 + 5c_2 - 4 - 3$$

Simplifying these equations gives

$$c_1 + c_2 = 3$$

$$c_1 + 5c_2 = 7$$

With $c_1 = 2$ and $c_2 = 1$, the general solution becomes

$$y(x) = 2e^x + e^{5x} - 2e^{2x} - e^{3x}$$

$$y(x) = 2e^x - 2e^{2x} - e^{3x} + e^{5x}$$

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Topic: Initial value problems with nonhomogeneous equations

Question: Solve the initial value problem for the second order nonhomogeneous equation, given $y(0) = 5$ and $y'(0) = 7$.

$$y'' - 2y' + 10y = 100x$$

Answer choices:

A $y(x) = e^x \cos(3x) + 4e^x \sin(3x) + 20x + 4$

B $y(x) = 2e^x \cos(3x) + e^x \sin(3x) + 20x + 4$

C $y(x) = 3e^x \cos(3x) - 2e^x \sin(3x) + 10x + 2$

D $y(x) = 4e^x \cos(3x) - 5e^x \sin(3x) + 10x + 2$

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Solution: C

First we solve the associated homogeneous equation.

$$y'' - 2y' + 10y = 0$$

$$r^2 - 2r + 10 = 0$$

$$(r-1)^2 + 9 = 0$$

$$r = 1 \pm 3i$$

So the complementary solution with complex conjugate roots is

$$y_c(x) = c_1 e^x \cos(3x) + c_2 e^x \sin(3x)$$

Our guess for the particular solution will be

$$y_p(x) = Ax + B$$

$$y_p'(x) = A$$

$$y_p''(x) = 0$$

Substituting into the original differential equation gives

$$0 - 2A + 10(Ax + B) = 100x$$

$$-2A + 10Ax + 10B = 100x$$

Equating coefficients gives $A = 10$ and $B = 2$, and therefore

$$y_p(x) = 10x + 2$$

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The general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1 e^x \cos(3x) + c_2 e^x \sin(3x) + 10x + 2$$

Differentiating the general solution gives

$$y'(x) = e^x(c_1 \cos(3x) + c_2 \sin(3x)) + e^x(-3c_1 \sin(3x) + 3c_2 \cos(3x)) + 10$$

Substitute the initial conditions $y(0) = 5$ and $y'(0) = 7$ into the general solution and its derivative.

$$c_1 + 0 + 0 + 2 = 5$$

$$c_1 + 3c_2 + 10 = 7$$

Simplifying these equations gives

$$c_1 = 3$$

$$c_2 = -2$$

With $c_1 = 3$ and $c_2 = -2$, the general solution becomes

$$y(x) = 3e^x \cos(3x) - 2e^x \sin(3x) + 10x + 2$$

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